

Aly Sibak

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EDUCATION

Bachelor of Computing (Honours), Computer Science (Co-op) 2023 – Present
University of Guelph — GPA: 83% — Area of Application: Project Management — **Dean's Honours List** (x2)

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy), TypeScript, JavaScript, SQL, Java, HTML5, CSS3, Bash

Frontend: React, Next.js, Tailwind CSS, Material-UI, Recharts, Zustand

Backend: Node.js, Express, RESTful APIs, PostgreSQL, SQLite

Cloud & DevOps: AWS (Lambda, S3, CloudFormation, API Gateway, Cognito, RDS, CloudWatch), Docker, Linux, CI/CD

AI / ML: Google Gemini API, prompt engineering, LLM behavioral metrics, experiment design, statistical analysis

Tools: Git, Jira, Bitbucket, Figma, Plaid API, Stripe, Winston

WORK EXPERIENCE

Software Developer Co-op May 2025 – Dec 2025
P&P Optica, Waterloo, ON

- Built full-stack **PPO Insights** dashboard using **React**, **TypeScript**, and **PostgreSQL**, serving **20+ facilities** across 3 continents (Tyson, Costco, Hormel) and tracking **1,000+ daily incidents** for food safety decisions
- Integrated **Google Gemini API** across **15+ relational tables**; ran repeated experiments across prompt configurations, measuring output consistency, latency, and task success across **1,000+ daily queries**; built a **Python/Pandas** pipeline processing **50MB / 10,000+ rows**, cutting bulk editing from **3 days to 5 minutes** and surfacing anomalies in data quality
- Automated **AWS CloudFormation/SAM** deployments across **9+ services** (Lambda, S3, RDS, API Gateway), reducing release time by **75%** (4 hrs → 1 hr); optimized **PostgreSQL** with **15 indexes**, improving query speed by **70%**
- Fixed a critical **S3 URL expiration bug** extending file access from **1 to 12 hours**, eliminating **200+ monthly support tickets**

Teaching Assistant — Web Design & Discrete Structures

Jan 2026 – Present

University of Guelph, Guelph, ON

- Support **80+ students** across two courses; run **3 hrs/week** office hours and provide written feedback on HTML5, CSS3, JavaScript ES6+, responsive design, and discrete mathematics

TECHNICAL PROJECTS

CarInfo — Car Comparison Platform  2024 – Present

React, TypeScript, Node.js, Express, PostgreSQL, Tailwind CSS, NHTSA API, Zustand, Recharts

- Full-stack platform covering **15,470 vehicles from 31 manufacturers (1995–2025)**; supports comparison of up to **5 vehicles** across **20+ specifications** with advanced filtering and a responsive dark theme
- Built **8 sorting algorithms** (Best Value, Bang-for-Buck, Lowest TCO), a **Value Matrix** scatter plot visualizing the full dataset, and a **TCO Calculator** with 5-year cost modeling
- Designed PostgreSQL schema with indexing, achieving **70% faster query response** across a large dataset with missing and inconsistent records

PocketChange — Spare Change Donation App 

2024 – Present

React, TypeScript, Node.js, Express, PostgreSQL, Plaid API, Stripe, Helmet.js, Winston, Zod, JWT

- Fintech app that rounds up transactions and donates spare change; integrated **Plaid API** for live bank connectivity and **Stripe** for donation processing to **40 charities** across **9 categories**
- Implemented rate limiting (**100 req/15 min**), **Zod** validation, **JWT** auth, OAuth, and Helmet.js; optimized database with **15 indexes** improving query speed by **60%**

ePortfolio — Investment Manager 

2024

Java, Object-Oriented Programming

- Console app managing **100+ investment holdings** (stocks and mutual funds) with buy/sell operations, price updates, gain calculations, and multi-criteria search by symbol, name, or price range
- Applied core OOP principles: inheritance, encapsulation, and polymorphism across a clean class hierarchy

LEADERSHIP & ACTIVITIES

Google Developer Student Club

2023 – Present

- Participate in hackathons and technical workshops on web development, API integration, and cloud architecture

Society of Computing and Information Science (SOCIS)

2023 – Present

- Mentor first-year students; organize technical workshops for **50+ CS students** on React, TypeScript, and REST APIs